

The Identification of a Second Lunar Calendar on Rongorongo Tablet C (Mamari)

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Fig 1. Tablet C 'Mamari'

Abstract

The author identifies a second lunar calendar on Rongorongo Tablet C, also known as the Mamari Tablet.

Author biography

Peter Berrett (ORCID: 0009-0008-7108-755X) is a Melbourne-based independent researcher with 75+ SOHO sungrazer discoveries including the first comet in Parker Solar Probe imagery.

1. Introduction

Rongorongo remains one of the world's few undeciphered scripts. Traditional approaches have focused on visual symbolism, narrative interpretation, or speculative linguistics. With the exception of the identification of what is believed to be a lunar calendar on Mamari Tablet C in 1958 by Thomas Barthel¹ and clarification of that calendar by Jacques Guy in 1990², progress of decipherment has been sparse.

2. Statement of Artificial Intelligence (AI) Use

While AI was occasionally consulted for historical research, to refine the presentation of this paper and for publishing assistance, any interpretive outputs were treated as nonbinding suggestions. The author's final analysis and conclusions are as demonstrated below and AI was not used in the pattern recognition aspects or conclusions of this paper.

¹ T. S. Barthel, *Grundlagen zur Entzifferung der Osterinselschrift* (Hamburg: Cram, de Gruyter & Co., 1958)

² "The Lunar Calendar of Tablet Mamari" (Journal de la Société des Océanistes, 1990, vol. 91, no. 2, pp. 135–149)

3. Analysis and discussion

Discovery of the second calendar, step by step

The following describes how the discovery of the second lunar calendar was made.

The transcription of the recto of Tablet C by Barthel³ is reproduced at Annexure A. The image of the recto was loaded into Microsoft Paint (Paint) and the author proceeded to cut and paste using Paint's copy tool. Paint is a simple program but at least it does not hallucinate like AI.

The first step undertaken was to take all the glyphs after the 3rd glyph on line 9 and move them down a considerable distance to allow for later work. Next, several delimiters and delimiter sequences were noticed and used to construct the chart as seen at Annexure B. This is not new work by the author but largely a reconstruction of Barthel's⁴ lunar calendar as clarified by Guy⁵. A correction has been made in light of a scribal correction identified by Horley⁶.

The structure of the lunar calendar is a recurring delimiter glyph sequence followed by a series of mostly moon inspired glyphs, crescent shaped and full moon shaped. At the end of each delimiter sequence is a composite glyph with a line coming out from the preceding glyph and down to a fish, much like a fishing line hauling up a newly caught fish. Up until the full moon (first 4 sequences) the fish is pointing upwards and downwards thereafter. It is observed that the fish pointing upwards aligns with the waxing of the moon and the fish pointing downwards with the waning.

³ T. S. Barthel, 'Mamari Tablet (Text C) Recto Line 6-9', *Wikimedia Commons*, https://commons.wikimedia.org/wiki/File:Barthel_Ca.png [accessed 4 April 2026]

⁴ Barthel, *Grundlagen*; see also 'Mamari Tablet (Text C) Recto Line 6-9', *Wikimedia Commons*

⁵ "The Lunar Calendar of Tablet Mamari" (*Journal de la Société des Océanistes*, 1990, vol. 91, no. 2, pp. 135–149)

⁶ Paul Horley, 'Rongorongo Script: Carving Techniques and Scribal Corrections', *Journal de la Société des Océanistes*, 129 (2009), 254

The summation of moon-shaped glyphs across the eight sequences yields a total of 31 ($2+6+4+4+5+3+5+2= 31$) moon shaped glyphs (moon nights). The glyphs following the 3rd delimiter sequence might be 3 moons or 4 but that question will be addressed later. It suffices to say that this is essentially the lunar calendar identified by Barthel⁷.

There remain 6 lines of glyphs, most of which include peculiar ‘poles’ of 3 diamonds, some with nodules attached to either side. These glyphs may represent a numerical or counting system; however, the underlying structure of that system is not yet understood. For the purposes of this paper, the stack of diamonds will be referred to as a staff and the nodules as buds. The number of diamonds in a staff varies but staves are always vertical poles.

Looking through the glyphs and disregarding the ‘numbers’ the author noticed a repeating sequence of glyphs (predominantly the same and repeating) and the staff & bud glyphs appeared to follow these in the same way as the moon crescents followed the delimiter sequences in the previously identified lunar calendar (A). The pattern is similar:

Delimiter sequence – staves/buds - delimiter sequence – staves/buds... etc

Following the pattern of Lunar Calendar A, the author respaced the glyphs as shown in Annexure C (latter half of Mamari Tablet recto). As readers will note, there is again a pattern⁸ consisting of a delimiter glyph sequence followed by some glyphs largely consisting of mixed staves with buds or no buds. This block has been titled ‘Lunar Calendar B’. The delimiter sequence generally has 5 glyphs and ends with a character that looks like a hooded anthropomorphic figure.

⁷ Barthel, *Grundlagen*, 42

⁸ After the author’s own independent identification, it was noted that the sequence was highlighted by user Kwamikagami in a Wikimedia image. While Kwamikagami’s image does not identify this as a lunar calendar and the bud count appears inconsistent with the present analysis, the digital trace provides useful context. See *Rongorongo Tablet C (Mamari)*, recto, lines Ca6–Ca9; digital trace and contrast adjustments by Kwamikagami, "File:Rongorongo C-a Mamari calendar.jpg," Wikimedia Commons, https://en.wikipedia.org/wiki/Rongorongo_text_C#/media/File:Rongorongo_C-a_Mamari_calendar.jpg (accessed May 17, 2026), based on photographic plates from Stéphen-Chauvet, *L’île de Pâques et ses mystères* (Paris: Éditions Tel, 1935).

The next step was to take all of block B and move it next to Lunar Calendar A (the calendar identified by Barthel⁹). Note that it has been aligned one row down. The eighth delimiter sequence for Lunar Calendar A has now been moved down to create more room for Delimiter sequence 7. The two aligned calendars are now shown at Annexure E.

As can be seen, there appears to be a common sequence of 6,4,4,5,3,5 then 2. This pattern is true for both the sequence of nights in Lunar Calendar A and the sequence of buds in Lunar Calendar B, subject to the following.

In each delimiter sequence the furry staff connected to or next to what appears to be a 4 pronged turtle appears to be the primary delimiter but it disappears after sequence 5 and only the second glyph (the second glyph in the delimiter sequence) appears in sequences 6a, 6b and 6c. This suggests that the first glyph is acting as a chapter whereas the second glyph denotes a subsection of a chapter.

In Lunar Calendar B, sequence 6, there is therefore:

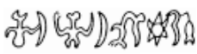
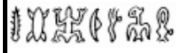
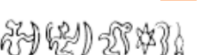
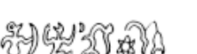
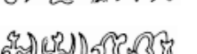
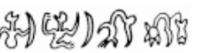
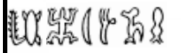
1. A secondary delimiter with **2** staves of 3 diamonds (sequence 6a) then;
2. A secondary delimiter with **2** staves of 3 diamonds (sequence 6b) then;
3. A secondary delimiter with **1** staff of 4 diamonds (sequence 6c)

To reinforce the above, all the buds on glyphs 6a, 6b and 6c add up to 5 which matches the sequence of 5 nights in the Lunar Calendar A cycle (7th sequence).

An argument could be made that the two sequences do not in fact match because one has 5 nights in one sequence and {2,2,1} nights in the other. That is a perfectly reasonable argument to make. In rebuttal the author contends that the 5 nights divided into 3 parcels represents the allocation of time (nights) to a particular task, or as having some feature peculiar to the purpose of Lunar Calendar B but that is not relevant to the purpose of Lunar Calendar A. It is further submitted that the truncated delimiter sequence for the {2,2,1} nights indicates a segmenting of a period of time and that the truncated delimiter sequence communicates a commonality between the 5 nights divided into 3 parcels.

⁹ Barthel, *Grundlagen*, 42

Fig 2. Comparison of calendars based on Barthel's drawings

Lunar calendar A				Lunar Calendar B			
			2			6	
			6			4	
			4			4	
			4			4	
			5			5	
			3			3	
			5			1	
						2	
						2	5
			2			2	
Total nights = 31				Total buds = 29			

1. This is a scribal correction identified by Paul Horley

The common sequences of the two calendars are unlikely to have arisen by chance, particularly given that they occur on the same side of the same tablet and share a structure of alternating delimiters and numerical items. This is a structural argument, not a semantic decipherment.

If, for example, Lunar Calendar B reflects an agricultural cycle, the buds may represent days rather than nights. In any case, it appears to follow the same underlying structural pattern as Lunar Calendar A.

It is noted that the buds appear on either side of the glyphs so this may represent mornings and afternoons but this is conjecture. No ethnographic source confirms this.

Lunar Calendar B is based on the same lunar calendar used by the Rapanui, the total buds add up to 29 which is just short of the visible lunar cycle of 29.53 days. In view of this, a Rapanui lunar calendar of 29 days seems more plausible than a calendar of 31 days. The sequences appear to be aligned with one another, and on this basis the second sequence of delimiters is interpreted as representing a related calendrical structure (henceforth referred to as Lunar Calendar B). The question is, what do the extra two days in Lunar Calendar A represent?

What follows is merely conjecture, not proof of anything, but it is theorised that the Rapanui had a 29 day calendar with an extra day added every second month. This would give a 29.5 day cycle which is just short of the actual lunar cycle length. In Lunar Calendar A there are two seated figures back to back next to the turtle in the final sequence of glyphs. It is suggested that the back to back characters may represent 'alternating'. Some support for this theory can be found in the 5th sequence of Lunar Calendar A. In most of the sequences, the third delimiter sequence glyph is some kind of bird facing to the right. However, immediately following the full moon the glyph faces to the left and thereafter reverts back to the right. It suggests that a change of some kind has occurred.

Guy¹⁰ compared the calendars obtained by Thomson (1889), Englert (1948) and Métraux (1940). One variance in the calendars is that Englert's calendar has 7 nights starting after the 6 night numbered sequence and ending with the full moon glyph in Lunar Calendar A. However, Thomson and Métraux identify an extra night thus giving 8 nights not 7.

Each of the authors identifies 2 nights before the six night numbered sequence but these are the darkened nights when the moon is in earth's shadow. They do not appear in Lunar Calendar B and may serve another purpose.

From the perspective of an observer, it does not make sense to start counting moons until one can see a moon or at least part of a crescent. It is difficult to count something that has not appeared yet. It is no surprise then, that the first six visible moons are numbered 1-6 by the Rapanui and this is agreed in each of the calendars. It also makes adjusting the cycle to match up with the observed moons much simpler if one waits until the moon is dark and then starts the next observing cycle when the first sign of a new crescent is observed.

¹⁰ Jacques B. M. Guy, 'On the Lunar Calendar of Tablet Mamari', *Journal de la Société des Océanistes*, 91.2 (1990), 135-49 (pp. 140-42).

The above assertion aligns with the structure of the Lunar Calendar B. It starts on day 1 (Kokore Tahi) and according to the number of buds has a 29 day cycle. It is possible that the glyphs preceding Lunar Calendar A or in the final delimiter sequence of Lunar Calendar A (or both) may contain instructions for adding a night every second month.

A structurally similar sequence

A considerable number of glyphs preceding the Lunar Calendar A block exist and these were examined to look for delimiting sequences.

The result 'The Preceding Glyphs Pattern' can be found at Annexure F.

A short recurring delimiter sequence (predominantly two glyphs) is observed elsewhere on the recto; however, unlike the two lunar calendars, the author sees nothing calendrical in that structure.

It is noted that the sequence of a turtle followed by back to back seated figures (boxed in red) is identical to that found in the glyphs in Lunar Calendar A, sequence 8. Put another way that sequence including the back to back characters is found close before the first 2 moons in Lunar Calendar A and immediately before the final 2 moons in Lunar Calendar A.

4. Transcription testing

The above analysis based on Barthel's¹¹ transcriptions was relatively straightforward. However, a complication emerged. The author is grateful for feedback from Jan Rich-Frel who suggested it would be prudent to check that the Barthel's transcriptions matched that on the Mamari tablet. A search online quickly established that the published photographs of the tablet were low resolution and/or incorporated into pdf files.

However, the European Commission funded INSCRIBE project has laser scanned the Mamari Tablet and made a 3D scan and visualisation tool available¹² which facilitates close inspection of the relevant glyphs. The author wishes to express his sincere thanks to the EU and INSCRIBE project members for what is an amazingly useful tool.

¹¹ Barthel, *Grundlagen*, 42

¹² "3D Interactive Web Viewer," *INSCRIBE: Invention of Scripts and Their Beginnings*, ERC INSCRIBE Project, Alma Mater Studiorum – University of Bologna, accessed May 13, 2026, https://www.inscribercproject.com/3d_viewer_home.php.

The author has analysed the sequences of glyphs and recounted the buds. Annexure G contains snapshots of each staff/bud glyph and his count of the buds on it. The numbers differ slightly from Barthel's¹³ but speculation will not change the final result. The question is – is this fatal to the author's conclusion?

Before answering that question one needs to step back and ask – what is the purpose of a lunar calendar? Lunar Calendar A is unlikely to exist just for the purpose of naming periods of time within the lunar month. The delimiter sequences are largely similar to each other and if one was naming periods of time like the 6 Kokore nights, one would expect them to be a range of different names as represented by different glyphs. However, they are largely the same with some minor variations.

The structure of Lunar Calendar B is similar to that of Lunar Calendar A. It exists on the same tablet on the same side (recto) as Lunar Calendar A and repeating glyphs feature (buds) that repeat in the same way as Lunar Calendar A (i.e delimiter – buds - delimiter-buds etc). The total number of buds adds up to 29 or 30 buds which is approximately the length of a lunar cycle.

The pattern of the number of buds, whilst not precisely the same as the pattern of the moons, is nevertheless very similar. The numbers of buds, as shown in Annexure G, often align with the number of moons at similar points in each respective sequence.

¹³ Barthel, *Grundlagen*, 42

Fig 3. Calendar comparison after analysis of INSCRIBE images

Barthel's Transcriptions			Per laser photos		Lunar Calendar A	
					2	
6			6		6	
4			4		4	
4			4 possibly 5		4	
5			4		5	
3			3		3	
1			1			
2			2			
2			3	6	5	
2			2		2	
Total = 29			Total = 29 or 30		Total = 30	

That there should be some variation in two lunar calendars should not be surprising. If there are two calendars for two different purposes e.g. fishing, agriculture, then two calendars could easily have different fishing and planting periods. Such is the case in the Maramataka¹⁴, the Maori lunar calendar. The best times for fishing differ from the best times for planting. However, the agricultural¹⁵ and fishing allocations of time fall within a common lunar cycle. Also, the Maramataka demonstrates that a lunar calendar can organise agricultural daytime activity, which would explain why Calendar B features plant-like bud glyphs rather than moon glyphs while still operating within a lunar cycle. The Rapanui and Māori calendars diverged for centuries, so parallels are suggestive, not determinative.

It is worth noting that Barthel's reconstruction does not exactly match the ethnographic calendars recorded by Thomson, Englert, and Métraux¹⁶. Their accounts differ in the number and placement of nights, yet Barthel's reconstruction is accepted because its structural features correspond plausibly to a lunar month. Calendar B should therefore be assessed by the same structural criteria, rather than by a demand for exact correspondence.

5. Conclusions

The sequences identified in the latter half of the recto of Tablet C have a similar structure as Lunar Calendar A and the pattern of buds in these sequences have a similar structure, albeit not exactly identical, to the pattern of nights in Lunar Calendar A. Therefore it is concluded that it is more probable than not that these sequences form a second lunar calendar.

Lunar Calendar A has sequences involving moons and fish whereas Lunar Calendar B has 'buds' and the first 6 buds are attached to what appear to be plant glyphs which is suggestive of an agricultural purpose. If the two calendars have different purposes then the allocation of time to each need not be exactly the same.

¹⁴ Te Papa (n.d.). 'Nights in the Maramataka | the Māori lunar month'. Available at: <https://www.tepapa.govt.nz/discover-collections/read-watch-play/maori/matariki-maori-new-year/nights-maramataka-maori-lunar> (Accessed: 12 May 2026).

¹⁵ Paul Horley (2011). "Lunar calendar in rongorongo texts and rock art of Easter Island." *Journal de la Société des Océanistes*, 132.

¹⁶ Jacques B. M. Guy, 'On the Lunar Calendar of Tablet Mamari', *Journal de la Société des Océanistes*, 91.2 (1990), 135-49 (p. 137) https://www.persee.fr/doc/jso_0300-953x_1990_num_91_2_2882.

The total of 29-30 total buds suggests that the Lunar Calendar B was either 29 or 30 nights long, commencing on the first sighting of the new moon. In the case of a 29 day lunar cycle, the addition of a day every 2 lunar cycles would more neatly allow the Rapanui to align their calendar with the actual visual lunar cycle however, it is not obvious how that extra half day would be accounted for in Lunar Calendar B. There is a range of possible explanations including that after 29 days the Rapanui just waited until they saw the new moon and then started again from there. Then again, the alternating back to back glyphs suggest that an extra night was added every two lunar cycles.

If the lunar cycle per Lunar Calendar B was 30 days long then that aligns with the 30 nights of Lunar Calendar A. Either interpretation suggests a lunar calendrical foundation.

The absence of an explicit intercalation mechanism in Lunar Calendar B need not be fatal to its interpretation as a lunar-based calendrical structure. A practical observational calendar tied to the first visible crescent may not require formal arithmetic correction. If the sequence allocated activities across approximately 29 days, the cycle could simply recommence upon the next visible new moon. Under such a system, occasional variation between the calendrical sequence and the precise astronomical lunar cycle would be operationally manageable without requiring a dedicated intercalary glyph or encoded correction system

Finally a sequence of glyphs with a similar structure to the two calendars is identified in the first 5 lines of the recto side of tablet C but no alignment with the lunar calendars is noted.

Notwithstanding the author's best effort to count the number of buds, the academic world would benefit from a professional reexamination of Tablet C to form a considered opinion as to why the stave and bud glyphs differ from that drawn by Barthel and to arrive at a more forensic count of the number of buds.

6. Author Note

Peter Berrett is an independent researcher.

7. Annexures

Annexure A Tablet C (Mamari) Recto

Line

[illegible]

Annexure B - Reordered Tablet C (Recto)

[illegible]

11111
 22222
 33333
 44444

ቶ(ቶ)ድቶ		ሀ ፕ
ቶ(ቲ) 		((((
ቶ(ቲ) ድቶ		(ፀፀ
ቶቲድቶ		ሀ(ፀፀ
ቶ(ቲ)ጽቶ		(((((
ቶ(ቲ)ድቶ		((ፕ
ቶ(ቲ)ድቶ	ሹ	(((((
ቶ(ቲ)ድቶ	ቶፊኒ	((

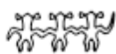
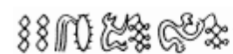
1. This is a scribal correction identified by Paul Horley

[illegible]

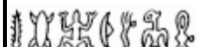
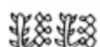
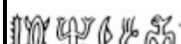
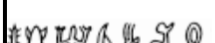
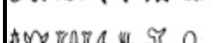
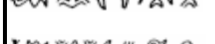
Annexure C - Lunar Calendar B

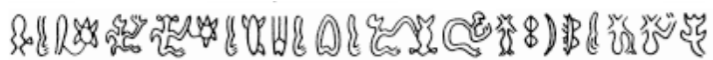




Lunar Calendar B





[illegible]

0000
 1111
 2222
 3333

[illegible][illegible]

ಕುಲದಿಗುಣಃ (೪) ಪುನಃ ಪುನಃ

Annexure E - Numbering of the two lunar calendars

Lunar calendar A			
ታላቁን ጥቅም	ሀገር	2	
ታላቁን ጥቅም	ሀገር	6	
ታላቁን ጥቅም	ሀገር	4	
ታላቁን ጥቅም	ሀገር	4	
ታላቁን ጥቅም	ሀገር	5	
ታላቁን ጥቅም	ሀገር	3	
ታላቁን ጥቅም	ሀገር	5	
ታላቁን ጥቅም	ሀገር	2	
Total nights = 31			

Lunar Calendar B			
ጥቅም	ጥቅም	6	
ጥቅም	ጥቅም	4	
ጥቅም	ጥቅም	4	
ጥቅም	ጥቅም	5	
ጥቅም	ጥቅም	3	
ጥቅም	ጥቅም	1	
ጥቅም	ጥቅም	2	
ጥቅም	ጥቅም	2	5
ጥቅም	ጥቅም	2	2
Total buds = 29			

1. This is a scribal correction identified by Paul Horley

Annexure F - Reordered preceding glyphs (First 5 lines of Tablet C)

[illegible]

အထွေထွေ အချက်အလက်

ਖੁਭ	ਮ
ਖਾੜਖ	ਨ
ਖਾਲ	ਨਿ
ਖਾਲੇ	ਓਲੇ
ਖਾਲੇ	ਫੈ
ਖਾਲੇ	ਘੋ
ਖਾਲੇ	ਗਿਲੇ

[illegible]

Annexure G - Transcription comparison with tablet glyphs

Barthel's Transcriptions			Per laser photos		Lunar Calendar A
					2
6			6		6
4			4		4
4			4 possibly 5		4
5			4		5
3			3		3
1			1		
2			2		
2			3	6	5
2			2		2
Total = 29			Total = 29 or 30		Total = 30

8. Acknowledgements

The author gratefully acknowledges the INSCRIBE project's 3D visualization tool (University of Bologna), Jan Rich-Frel's transcription and other feedback, and Microsoft Paint's hallucination-free glyph reordering. No AI contributed to core analysis. Finally, I would like to thank my cat, Henry, for constant companionship during the preparation of this manuscript.

9. References

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